Dhruba Dutta Chowdhury

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RESEARCH INTERESTS

Black Holes, Dark Matter, Galaxies, Gas and Stellar Dynamics, Star Clusters

POSITIONS

Center for Astrophysics and Supercomputing,
Swinburne University of Technology, Melbourne, Australia Incoming

- Postdoctoral Research Associate
- Advisor: Darren Croton

Racah Institute of Physics,
The Hebrew University of Jerusalem, Jerusalem, Israel Oct 2022-Sep 2025

- Postdoctoral Fellow (Oct 2024-Sep 2025)
- Israel Academy of Sciences and Humanities Postdoctoral Fellow (Oct 2022-Sep 2024)
- Advisors: Avishai Dekel and Nir Mandelker

Department of Physics,
Presidency University, Kolkata, India Sep 2015-Aug 2016

- Project Assistant (Junior Research Fellow)
- Project: Modeling the 21 cm Signal from the Dark Ages
- Advisor: Kanan Kumar Datta

EDUCATION

Yale University, New Haven, USA Aug 2016-Aug 2022

- Ph.D., M.S., M.Phil. in Astronomy
- Thesis: Constraining Dark Matter through Gravitational Heating and Cooling Processes
- Advisors: Frank van den Bosch and Pieter van Dokkum

Presidency University, Kolkata, India Aug 2013-Aug 2015

- M.Sc. in Physics
- Thesis: The Sunyaev-Zel'dovich Signal from Quasar Host Halos
- Advisor: Suchetana Chatterjee

Presidency College, University of Calcutta, Kolkata, India July 2010-July 2013

- B.Sc. (Honors) in Physics
- Minors in Mathematics and Chemistry

AWARDS

- Arnold Rosenblum Prize for Excellence in Research in Astrophysics, Hebrew University 2023
- Israel Academy of Sciences and Humanities Postdoctoral Fellowship 2022-2024
- Sheldon Wise Pre-Doctoral Fellowship, Yale University 2017-2018
- Junior Research Fellowship, Department of Science & Technology, India 2015-2016
- Lilabati Ray Memorial Prize for Best Student Seminar, Presidency University 2015
- INSPIRE scholarship, Department of Science & Technology, India 2010-2015

PROFESSIONAL ACTIVITIES

- Referee for The Astrophysical Journal — 2019-Present
- Referee for Classical and Quantum Gravity — 2025-Present
- Astro-ph Meeting Moderator, The Hebrew University of Jerusalem — 2022-Present
- Yale Astronomy Graduate Student Talks SOC Member — Spring 2019
- Galaxy Lunch Moderator, Yale Astronomy Department — 2017-2018

TEACHING EXPERIENCE

- Guest Lecturer, Advanced Astrophysics: Galaxies and Cosmology, Hebrew University — Spring 2023, 2024
- Teaching Fellow, Planets and Stars, Yale University — Spring 2017
- Teaching Fellow, Galaxies and the Universe, Yale University — Fall 2017, 2019

CONFERENCE TALKS

1. Galaxy Formation Workshop, University of California, Santa Cruz — Aug 2025
   Growth of Massive Black Holes in FFB Disks at Cosmic Dawn
2. Cosmic Dawn Revealed by JWST: First Stars, Galaxies, and Black Holes, KITP, Santa Barbara — Aug 2024
   Dynamical Friction in Disks and Application to FFB Disks
3. Galaxy Formation Workshop, University of California, Santa Cruz — Aug 2024
   Radial Transport in High-Redshift Disks
4. Galaxy Formation Workshop, University of California, Santa Cruz — Aug 2023
   Radial Transport in High-Redshift Disks
5. 68th Israel Physical Society Meeting, Tel Aviv — April 2023
   Radial Transport in High-Redshift Disks
6. Galaxy Formation Workshop, University of California, Santa Cruz — Aug 2022
   Constraining Dark Matter with Gravitational Heating and Cooling Processes
7. Virtual Workshop on Very Light Dark Matter, Kavli IPMU, Kashiwa — Sept 2021
   On the Random Motion of Nuclear Objects in a Fuzzy Dark Matter Halo
8. Virtual Young Astronomers on Galactic Nuclei Meeting, Niels Bohr Institute, Copenhagen — Sept 2021
   On the Random Motion of Nuclear Objects in a Fuzzy Dark Matter Halo
9. Virtual 16th Marcel Grossmann Meeting — July 2021
   On the Random Motion of Nuclear Objects in a Fuzzy Dark Matter Halo
10. Virtual 238th American Astronomical Society Meeting — June 2021
    On the Random Motion of Nuclear Objects in a Fuzzy Dark Matter Halo
11. Epoch of Reionization Workshop, Indian Institute of Technology, Kharagpur — July 2016
    Imprints of the Recombination History of the Universe on the 21-cm Signal from the Dark Ages
12. Topical Conference on Gravity, Cosmology, Astronomy, and Astrophysics, IISER, Kolkata — Sept 2015
    Sunyaev–Zel'dovich Signal from Quasar Hosts: Implications for Quasar Feedback Detection

SEMINARS

1. Fortnightly Meeting, Center for Dark Matter Particle Physics, Melbourne, Australia — Sep 2026
2. State of the Universe Seminar, Tata Institute of Fundamental Research, India (**invited**) — Apr 2023
3. Nature of Dark Matter on Small Scales Virtual Seminar (**invited**) — Apr 2022
4. Galaxies and Cosmology Seminar, University of Texas at Austin, USA (**invited**) — Nov 2021
5. Galaxy Coffee Talk, Max Planck Institute for Astronomy, Germany — Nov 2021
6. Astro Lunch Seminar, Carnegie Mellon University, USA (**invited**) — Nov 2021
7. Cosmology Group Meeting Talk, Canadian Institute for Theoretical Astrophysics, Canada — Nov 2021
8. L2G2 Meeting Talk, Center for Computational Astrophysics, USA (**invited**) — Nov 2021
9. Lunch Talk, Leiden Observatory, Netherlands — Nov 2021
10. Lunch Talk, Carnegie Observatories, USA (**invited**) — Nov 2021
11. Center for Astrophysics Seminar, Harvard University, USA (**invited**) — Nov 2021
12. Cosmology Seminar, Max Planck Institute for Astrophysics, Germany — Oct 2021
13. Thunch Talk, Princeton University, USA — Oct 2021
14. CCAPP Seminar, Ohio State University, USA (**invited**) — Oct 2021

15. Flash Talk, University of California, Santa Cruz, USA — Oct 2021
16. Brown Bag Lunch Talk, Massachusetts Institute of Technology, USA — Oct 2021
17. TAPIR Seminar, California Institute of Technology, USA (**invited**) — Oct 2021
18. Cosmo Lunch Talk, The Hebrew University of Jerusalem, Israel (**invited**) — Sep 2021
19. Physics Club Talk, Presidency University, India (**invited**) — Jul 2019

INVITED COLLOQUIA

1. School of Astrophysics, Presidency University, Kolkata, India — Sept 2022
2. Academia Sinica Institute for Astronomy and Astrophysics, Taipei, Taiwan — Jan 2022
3. Department of Physics, Presidency University, Kolkata, India — May 2019

CONFERENCE POSTERS

1. Santa Cruz Galaxy Workshop, University of California, Santa Cruz — Aug 2019
   On the Orbital Decay of Globular Clusters in NGC 1052-DF2
2. Small Galaxies, Cosmic Questions Conference, Durham University, Durham — July 2019
   On the Orbital Decay of Globular Clusters in NGC 1052-DF2

PUBLICATIONS [ADS]

Total: 15, 433 citations, h-index = 10 (As on 9 Oct, 2026)

First Author

1. **Dutta Chowdhury, D.**, Dekel, A., Mandelker, N., Ginzburg, O., and Genzel, R. "Radial Transport in High-Redshift Disk Galaxies Dominated by Inflowing Streams" 2025, A&A, 704, A1
2. **Dutta Chowdhury, D.**, van den Bosch, F.C., van Dokkum, P., Robles, V.H., Schive, H. et al. "On the Dynamical Heating of Dwarf Galaxies in a Fuzzy Dark Matter Halo" 2023, ApJ, 949, 68
3. **Dutta Chowdhury, D.**, van den Bosch, F.C., Robles, V.H., van Dokkum, P., Schive, H. et al. "On the Random Motion of Nuclear Objects in a Fuzzy Dark Matter Halo" 2021, ApJ, 916, 27
4. **Dutta Chowdhury, D.**, van den Bosch, F.C., and van Dokkum, P. "On the Evolution of the Globular Cluster System in NGC 1052-DF2: Dynamical Friction, Globular-Globular Interactions, and Galactic Tides" 2020, ApJ, 903, 149
5. **Dutta Chowdhury, D.**, van den Bosch, F.C., and van Dokkum, P. "On the Orbital Decay of Globular Clusters in NGC 1052-DF2: Testing a Baryon Only Mass Model" 2019, ApJ, 877, 133
6. **Dutta Chowdhury, D.** and Chatterjee, S. "Sunyaev-Zel'dovich Signal from Quasar Hosts: Implications for Detection of Quasar Feedback" 2017, ApJ, 839, 34

Student Mentee Led

7. Bradley, M., Croton, D.J., Paun, R.A.M., **Dutta Chowdhury, D.** et al. "SAGE26 Paper I: Modeling the baryon cycle from cosmic dawn to the present day" 2026, arxiv:2610.09564
   (*helped develop the idea of incorporating feedback free bursts in SAGE and provided comprehensive feedback*)

Other Significant Contributions

8. Dekel, A., **Dutta Chowdhury, D.** et al. "From Feedback Free Star Clusters to Little Red Dots via Compaction" 2025, arXiv:2511.07578, accepted for publication in ApJ
   (*helped with project idea development, made selected plots, and contributed to writing*)
9. Dekel, A., Stone, N., **Dutta Chowdhury, D.** et al. "Growth of Massive Black Holes in FFB Galaxies at Cosmic Dawn" 2025, A&A, 695, A97
   (*ran and analyzed numerical simulations and contributed to writing*)
10. Ansar, S., Datta, K.K. and **Dutta Chowdhury, D.** "Impact of Inhomogeneous CMB Heating of Gas on the HI 21-cm Signal During Dark Ages" 2018, PhysRevD, 98, 103505
    (*initiated the study and did a portion of the analytical calculations*)

Further Collaborations

11. Dekel, A. et al. including **Dutta Chowdhury, D.** "From FFB Starbursts at Cosmic Dawn to Quenching at Cosmic Morning: High-z Galaxy Bimodality" 2025, MNRAS, 544, 1

12. Ginzburg, O. et al. including **Dutta Chowdhury, D.** "On the origin of compressive turbulence in proto-clumps in high redshift disks" 2025, A&A, 698, A110
13. van Dokkum, P. et al. including **Dutta Chowdhury, D.** "Monochromatic globular clusters as a critical test of formation models for the dark matter deficient galaxies NGC1052-DF2 and NGC1052-DF4" 2022, ApJL, 940, L9
14. van Dokkum, P. et al. including **Dutta Chowdhury, D.** "A trail of dark-matter-free galaxies from a bullet-dwarf collision" 2022, Nature, 605, 435
15. Shen, Z. et al. including **Dutta Chowdhury, D.** "A Tip of the Red Giant Branch Distance of 22.1±1.2 Mpc to the Dark Matter Deficient Galaxy NGC 1052–DF2 from 40 Orbits of Hubble Space Telescope Imaging" 2021, ApJL, 914, L12

Student Mentoring

1. Amrita Banerjee, PhD Student, Swinburne University of Technology 2026 - Present
   Topic: Semi-Analytical Modeling of Positive AGN Feedback (Advisor: Darren Croton)
2. Michael Bradley, PhD Student, Swinburne University of Technology 2025 - Present
   Topic: Semi-Analytical Modeling and High Redshift Galaxies (Advisor: Darren Croton)
3. Ryan Montesi, Undergraduate Student, Yale University 2019-2020
   Topic: Dynamical Friction in Cored vs Cuspy Dark Matter (Advisor: Frank van den Bosch)

COMPUTATIONAL SKILLS

- Collisionless N-Body simulations with GADGET and RAMSES
- Fuzzy Dark Matter simulations with GAMER-2 (AMR Code)
- Self Interacting Dark Matter simulations with GIZMO
- Programming skills in C, FORTRAN 77, and Python
- Post processing and analysis of Hydro-Cosmological Simulations

Software

- `SIDM-GTF`: A Python-based package for evolving SIDM halos by solving the coupled gravothermal fluid equations using a fully-implicit scheme.